

Land Value vs. Resident Welfare as the Municipal Objective

Setup

This note isolates a single question raised in §Municipal Zoning of `model.tex`: does maximizing local land value give the same zoning policy as maximizing incumbent-resident welfare, in this model’s absentee-landlord environment? Bordeu (2026) models the local government as maximizing land value and notes this is sometimes equivalent to maximizing aggregate resident welfare $\bar{U}$; this note makes precise *why that equivalence fails here*, and what assumption would need to change to restore it.

Notation follows `model.tex`: at location i , $\bar{U}_i = \Gamma(1 - \frac{1}{\epsilon})\Phi_i^{1/\epsilon}$ is the representative resident’s expected utility, N_i is population, q_i^R is the residential rent, $\bar{H}_i^R$ is the municipality’s chosen residential floor-space cap, and land at i is owned by *absentee* landlords who collect $q_i^R \bar{H}_i^R + q_i^F \bar{H}_i^F$ and consume it outside the metro (§Environment). The municipal welfare objective is $W^g = \sum_{i \in S^g} N_i \bar{U}_i$; define the land-value objective as $V^g = \sum_{i \in S^g} V_i$, $V_i \equiv q_i^R \bar{H}_i^R + q_i^F \bar{H}_i^F$.

Result

Proposition 1 (Non-equivalence under absentee landownership). *Fix all locations’ quantities $\{\bar{H}_i^R, \bar{H}_i^F\}_{i \in S^g}$ and consider a marginal increase in the residential rent q_i^R at a single location i driven by any force external to the quantity choice itself (e.g. a rise in aggregate housing demand at i). Then*

$$\frac{\partial V_i}{\partial q_i^R} = \bar{H}_i^R > 0, \quad \text{while} \quad \frac{d(N_i \bar{U}_i)}{dq_i^R} < 0.$$

Hence V^g and W^g are strictly increasing and strictly decreasing, respectively, in q_i^R : on the rent margin the two objectives point in opposite directions, so no monotone reparametrization of the model’s primitives $(\alpha, \epsilon, a, b, \eta, \gamma)$ can make land-value maximization and welfare maximization select the same zoning policy in general.

Proof. Land value. By definition, holding $\bar{H}_i^R$ fixed, $\partial V_i / \partial q_i^R = \bar{H}_i^R > 0$: a marginal rent increase raises land value one-for-one with the quantity of residential floor space it applies to.

Welfare. From the household block (§Households), Φ_i enters through $\tilde{V}_{ij} \propto (q_i^R)^{-(1-\alpha)}$, giving the exact partial already derived in §Municipal Zoning of `model.tex`,

$$\frac{\partial \bar{U}_i}{\partial q_i^R} = -\frac{(1-\alpha) \bar{U}_i}{q_i^R} < 0.$$

This is a pure price effect with *no offsetting income effect*: because landlords are absentee, the rent a resident of i pays leaves the local economy entirely and does not return to residents through any budget-constraint or amenity channel (§Environment). Population N_i is likewise weakly decreasing in q_i^R : a higher rent lowers $\bar{V}_{ij}$ for every j , hence lowers π_{ij} for every j (eq. `eq:pi` in `model.tex`), hence lowers $N_i = \sum_j N\pi_{ij}$. So

$$\frac{d(N_i \bar{U}_i)}{dq_i^R} = \underbrace{N_i \frac{\partial \bar{U}_i}{\partial q_i^R}}_{<0} + \underbrace{\bar{U}_i \frac{\partial N_i}{\partial q_i^R}}_{\leq 0} < 0.$$

Both terms are non-positive and the first is strictly negative, so the sum is strictly negative. ■ $\square$

Remark. The proof isolates the cleanest possible channel — a pure rent increase, quantities fixed — precisely so the sign conflict cannot be an artifact of the municipality’s specific instrument. It is structural: land value is high exactly when residents are paying a lot for housing, while welfare is high exactly when they are not. Under absentee ownership, what is a cost to the resident is not simultaneously a benefit to the resident acting in some other capacity (e.g. as landowner) — it is a pure leakage out of the metro.

Why land value is a fundamentally different objective, not just a differently weighted one

Proposition 1 shows the two objectives disagree at the margin; the corollary below shows they disagree in a more basic, structural sense — because V_i does not depend on the amenity/congestion terms that are the entire content of the welfare objective’s zoning trade-off in §Municipal Zoning.

Corollary 1.1 (Land value is congestion- and amenity-blind). *Using the residential rent-clearing condition (eq. `eq:rent-clearing` in `model.tex`), $q_i^R \bar{H}_i^R = (1 - \alpha) \sum_j N\pi_{ij} w_j$, so land value at i can be written purely in terms of income flows,*

$$V_i = (1 - \alpha) \sum_j N\pi_{ij} w_j + q_i^F \bar{H}_i^F,$$

with no dependence on $\bar{B}_i$, $\phi(N_i, L_i)$, or $\bar{U}_i$ itself. A land-value-maximizing municipality is therefore indifferent, holding aggregate resident income $\sum_j N\pi_{ij} w_j$ fixed, to how much of that income residents actually get to keep as utility versus pay out as rent, and it places no direct weight on the residential-crowding term $\phi(N_i, L_i)$ that drives channels (3)–(4) of `model.tex`’s zoning-foc-explicit. Welfare maximization weighs exactly these terms.

Remark (What would restore equivalence). Bordeu (2026)’s equivalence result requires residents to *be* the landowners who capture their own parcel’s rent. If §Environment’s absentee-landlord assumption were replaced with full local rebate — each resident of i receiving a $1/N_i$ share of $q_i^R \bar{H}_i^R + q_i^F \bar{H}_i^F$ as a lump-sum transfer added to the budget constraint — the price effect in Proposition 1 would be exactly offset by an equal wealth effect for residents who both pay and receive the rent, and $\partial \bar{U}_i / \partial q_i^R$ would no longer be signed by the price channel alone. Even then, equivalence with land-value maximization is not automatic: it requires the specific Cobb-Douglas/Fréchet functional forms here to

make the wealth effect cancel the price effect *exactly*, not just partially, which is a knife-edge property of the model rather than a generic one. This is left as a possible extension alongside the land-supplier extension already flagged in §Municipal Zoning of `model.tex` (item 13 of `CLAUDE.md`'s Open Model Questions).

References

- [1] Olivia Bordeu. *Commuting Infrastructure in Fragmented Cities*. 2026.